Dhirubhai Ambani University Technology

Formerly DA-IICT

Instagram-like Social Media Database

BCNF Normalization Proof Report

"USERS" Relation:

USERS(user_id, username, email, password_hash, created_at, is_verified, type_id)

FDs:

{user_id} → {username, email, password_hash, created_at, is_verified, type_id}

Closure of {user_id}:

{user_id}+ = {user_id, username, email, password_hash, created_at, is_verified, type_id}

Closure of {user_id} covers all the attributes of the Relation "USERS".
Thus {user_id} is a candidate key.

Candidate Key: {user_id}

Since Candidate key is the subset of Super Key

The determinant of all the functional dependencies of the relation "USERS" is a Super Key.
Thus given relation is in **BCNF (Boyce-Codd Normal Form)**

"ACCOUNT_TYPES" Relation:

ACCOUNT_TYPES(type_id, type_name)

FDs:

{type_id} → {type_name}

Closure of {type_id}:

{type_id}+ = {type_id, type_name}

Closure of {type_id} covers all the attributes of the Relation "ACCOUNT_TYPES".
Thus {type_id} is a candidate key.

Candidate Key: {type_id}

Since Candidate key is the subset of Super Key

The determinant of all the functional dependencies of the relation "ACCOUNT_TYPES" is a Super Key.
Thus given relation is in **BCNF (Boyce-Codd Normal Form)**

"FOLLOWERS" Relation:

FOLLOWERS(follower_id, following_id, created_at)

FDs:

{follower_id, following_id} → {created_at}

Closure of {follower_id, following_id}:

{follower_id, following_id}+ = {follower_id, following_id, created_at}

Closure of {follower_id, following_id} covers all the attributes of the Relation "FOLLOWERS".
Thus {follower_id, following_id} is a candidate key.

Candidate Key: {follower_id, following_id}

Since Candidate key is the subset of Super Key

The determinant of all the functional dependencies of the relation "FOLLOWERS" is a Super Key.
Thus given relation is in **BCNF (Boyce-Codd Normal Form)**

"BLOCKED_ACCOUNTS" Relation:

BLOCKED_ACCOUNTS(blocker_id, blocked_id, block_date)

FDs:

{blocker_id, blocked_id} → {block_date}

Closure of {blocker_id, blocked_id}:

{blocker_id, blocked_id}+ = {blocker_id, blocked_id, block_date}

Closure of {blocker_id, blocked_id} covers all the attributes of the Relation "BLOCKED_ACCOUNTS".
Thus {blocker_id, blocked_id} is a candidate key.

Candidate Key: {blocker_id, blocked_id}

Since Candidate key is the subset of Super Key

The determinant of all the functional dependencies of the relation "BLOCKED_ACCOUNTS" is a Super Key.
Thus given relation is in **BCNF (Boyce-Codd Normal Form)**

"CLOSE_FRIENDS_LIST" Relation:

CLOSE_FRIENDS_LIST(user_id, friend_id, added_at)

FDs:

{user_id, friend_id} → {added_at}

Closure of {user_id, friend_id}:

{user_id, friend_id}+ = {user_id, friend_id, added_at}

Closure of {user_id, friend_id} covers all the attributes of the Relation "CLOSE_FRIENDS_LIST".
Thus {user_id, friend_id} is a candidate key.

Candidate Key: {user_id, friend_id}

Since Candidate key is the subset of Super Key

The determinant of all the functional dependencies of the relation "CLOSE_FRIENDS_LIST" is a Super Key.
Thus given relation is in **BCNF (Boyce-Codd Normal Form)**

"POSTS" Relation:

POSTS(post_id, caption, location_name, user_id)

FDs:

{post_id} → {caption, location_name, user_id}

Closure of {post_id}:

{post_id}+ = {post_id, caption, location_name, user_id}

Closure of {post_id} covers all the attributes of the Relation "POSTS".
Thus {post_id} is a candidate key.

Candidate Key: {post_id}

Since Candidate key is the subset of Super Key

The determinant of all the functional dependencies of the relation "POSTS" is a Super Key.
Thus given relation is in **BCNF (Boyce-Codd Normal Form)**

"POST_MEDIA" Relation:

POST_MEDIA(media_id, post_id, media_url, position)

FDs:

{media_id} → {post_id, media_url, position}

Closure of {media_id}:

{media_id}+ = {media_id, post_id, media_url, position}

Closure of {media_id} covers all the attributes of the Relation "POST_MEDIA".
Thus {media_id} is a candidate key.

Candidate Key: {media_id}

Since Candidate key is the subset of Super Key

The determinant of all the functional dependencies of the relation "POST_MEDIA" is a Super Key.
Thus given relation is in **BCNF (Boyce-Codd Normal Form)**

"COMMENTS" Relation:

COMMENTS(comment_id, content, user_id, post_id, parent_comment_id)

FDs:

{comment_id} → {content, user_id, post_id, parent_comment_id}
Note: parent_comment_id is a self-referential FK to comment_id (nullable).

Closure of {comment_id}:

{comment_id}+ = {comment_id, content, user_id, post_id, parent_comment_id}

Closure of {comment_id} covers all the attributes of the Relation "COMMENTS".
Thus {comment_id} is a candidate key.

Candidate Key: {comment_id}

Since Candidate key is the subset of Super Key

The determinant of all the functional dependencies of the relation "COMMENTS" is a Super Key.
Thus given relation is in **BCNF (Boyce-Codd Normal Form)**

"TARGET_TYPE" Relation:

TARGET_TYPE(target_type_id, target_type_name)

FDs:

{target_type_id} → {target_type_name}

Closure of {target_type_id}:

{target_type_id}+ = {target_type_id, target_type_name}

Closure of {target_type_id} covers all the attributes of the Relation "TARGET_TYPE".
Thus {target_type_id} is a candidate key.

Candidate Key: {target_type_id}

Since Candidate key is the subset of Super Key

The determinant of all the functional dependencies of the relation "TARGET_TYPE" is a Super Key.
Thus given relation is in **BCNF (Boyce-Codd Normal Form)**

"LIKES" Relation:

LIKES(like_id, user_id, target_id, target_type_id)

FDs:

{like_id} → {user_id, target_id, target_type_id}
{user_id, target_id, target_type_id} → {like_id}   [UNIQUE constraint — alternate key]

Closure of {like_id}:

{like_id}+ = {like_id, user_id, target_id, target_type_id}

Closure of {like_id} covers all the attributes of the Relation "LIKES".
Thus {like_id} is a candidate key.

Closure of {user_id, target_id, target_type_id}:

{user_id, target_id, target_type_id}+ = {user_id, target_id, target_type_id, like_id}

Closure of {user_id, target_id, target_type_id} covers all the attributes of the Relation "LIKES".
Thus {user_id, target_id, target_type_id} is a candidate key.

Candidate Key: {like_id}

Since Candidate key is the subset of Super Key

The determinant of all the functional dependencies of the relation "LIKES" is a Super Key.
Thus given relation is in **BCNF (Boyce-Codd Normal Form)**

"REELS" Relation:

REELS(reel_id, video_url, music_id, total_views, is_remixable, user_id)

FDs:

{reel_id} → {video_url, music_id, total_views, is_remixable, user_id}

Closure of {reel_id}:

{reel_id}+ = {reel_id, video_url, music_id, total_views, is_remixable, user_id}

Closure of {reel_id} covers all the attributes of the Relation "REELS".
Thus {reel_id} is a candidate key.

Candidate Key: {reel_id}

Since Candidate key is the subset of Super Key

The determinant of all the functional dependencies of the relation "REELS" is a Super Key.
Thus given relation is in **BCNF (Boyce-Codd Normal Form)**

"MUSIC_LIBRARY" Relation:

MUSIC_LIBRARY(music_id, audio_url, song_title, artist_name, duration_sec)

FDs:

{music_id} → {audio_url, song_title, artist_name, duration_sec}

Closure of {music_id}:

{music_id}+ = {music_id, audio_url, song_title, artist_name, duration_sec}

Closure of {music_id} covers all the attributes of the Relation "MUSIC_LIBRARY".
Thus {music_id} is a candidate key.

Candidate Key: {music_id}

Since Candidate key is the subset of Super Key

The determinant of all the functional dependencies of the relation "MUSIC_LIBRARY" is a Super Key.
Thus given relation is in **BCNF (Boyce-Codd Normal Form)**

"STORIES" Relation:

STORIES(story_id, media_url, expires_at, audience_type_id, user_id)

FDs:

{story_id} → {media_url, expires_at, audience_type_id, user_id}

Closure of {story_id}:

{story_id}+ = {story_id, media_url, expires_at, audience_type_id, user_id}

Closure of {story_id} covers all the attributes of the Relation "STORIES".
Thus {story_id} is a candidate key.

Candidate Key: {story_id}

Since Candidate key is the subset of Super Key

The determinant of all the functional dependencies of the relation "STORIES" is a Super Key.
Thus given relation is in **BCNF (Boyce-Codd Normal Form)**

"STORY_VIEWS" Relation:

STORY_VIEWS(story_id, viewer_id, view_time)

FDs:

{story_id, viewer_id} → {view_time}

Closure of {story_id, viewer_id}:

{story_id, viewer_id}+ = {story_id, viewer_id, view_time}

Closure of {story_id, viewer_id} covers all the attributes of the Relation "STORY_VIEWS".
Thus {story_id, viewer_id} is a candidate key.

Candidate Key: {story_id, viewer_id}

Since Candidate key is the subset of Super Key

The determinant of all the functional dependencies of the relation "STORY_VIEWS" is a Super Key.
Thus given relation is in **BCNF (Boyce-Codd Normal Form)**

"AUDIENCE_TYPE" Relation:

AUDIENCE_TYPE(audience_type_id, audience_type_name)

FDs:

{audience_type_id} → {audience_type_name}

Closure of {audience_type_id}:

{audience_type_id}+ = {audience_type_id, audience_type_name}

Closure of {audience_type_id} covers all the attributes of the Relation "AUDIENCE_TYPE".
Thus {audience_type_id} is a candidate key.

Candidate Key: {audience_type_id}

Since Candidate key is the subset of Super Key

The determinant of all the functional dependencies of the relation "AUDIENCE_TYPE" is a Super Key.
Thus given relation is in **BCNF (Boyce-Codd Normal Form)**

"HIGHLIGHTS" Relation:

HIGHLIGHTS(highlight_id, name, user_id)

FDs:

{highlight_id} → {name, user_id}

Closure of {highlight_id}:
{highlight_id}+ = {highlight_id, name, user_id}

Closure of {highlight_id} covers all the attributes of the Relation "HIGHLIGHTS".
Thus {highlight_id} is a candidate key.

Candidate Key: {highlight_id}

Since Candidate key is the subset of Super Key

The determinant of all the functional dependencies of the relation "HIGHLIGHTS" is a Super Key.
Thus given relation is in **BCNF (Boyce-Codd Normal Form)**

"CHATS" Relation:

CHATS(chat_id, sender_id, receiver_id, message_content, send_at, is_read, message_type_id)

FDs:

{chat_id} → {sender_id, receiver_id, message_content, send_at, is_read, message_type_id}

Closure of {chat_id}:
{chat_id}+ = {chat_id, sender_id, receiver_id, message_content, send_at, is_read, message_type_id}

Closure of {chat_id} covers all the attributes of the Relation "CHATS".
Thus {chat_id} is a candidate key.

Candidate Key: {chat_id}

Since Candidate key is the subset of Super Key

The determinant of all the functional dependencies of the relation "CHATS" is a Super Key.
Thus given relation is in **BCNF (Boyce-Codd Normal Form)**

"MESSAGE_TYPES" Relation:

MESSAGE_TYPES(message_type_id, type_name)

FDs:

{message_type_id} → {type_name}

Closure of {message_type_id}:

{message_type_id}+ = {message_type_id, type_name}

Closure of {message_type_id} covers all the attributes of the Relation "MESSAGE_TYPES".
Thus {message_type_id} is a candidate key.

Candidate Key: {message_type_id}

Since Candidate key is the subset of Super Key

The determinant of all the functional dependencies of the relation "MESSAGE_TYPES" is a Super Key.
Thus given relation is in **BCNF (Boyce-Codd Normal Form)**

"CALLS" Relation:

CALLS(call_id, caller_id, receiver_id, started_at, duration_seconds, status, call_type_id)

FDs:

{call_id} → {caller_id, receiver_id, started_at, duration_seconds, status, call_type_id}

Closure of {call_id}:

{call_id}+ = {call_id, caller_id, receiver_id, started_at, duration_seconds, status, call_type_id}

Closure of {call_id} covers all the attributes of the Relation "CALLS".
Thus {call_id} is a candidate key.

Candidate Key: {call_id}

Since Candidate key is the subset of Super Key

The determinant of all the functional dependencies of the relation "CALLS" is a Super Key.
Thus given relation is in **BCNF (Boyce-Codd Normal Form)**

"CALL_TYPES" Relation:

CALL_TYPES(call_type_id, type_name)

FDs:

{call_type_id} → {type_name}

Closure of {call_type_id}:

{call_type_id}+ = {call_type_id, type_name}

Closure of {call_type_id} covers all the attributes of the Relation "CALL_TYPES".
Thus {call_type_id} is a candidate key.

Candidate Key: {call_type_id}

Since Candidate key is the subset of Super Key

The determinant of all the functional dependencies of the relation "CALL_TYPES" is a Super Key.
Thus given relation is in **BCNF (Boyce-Codd Normal Form)**

Final Conclusion

All the 20 relations in the database satisfy the BCNF condition.
For every non-trivial functional dependency $X \rightarrow Y$, X is a superkey.
Therefore, all relations are already in BCNF, and no decomposition was required.

This is because:
- Each entity has a well-defined primary key
- Alternate keys (like email and username) are handled properly
- Relationships involving multiple entities are handled using separate relations, which avoids hidden dependencies

So overall, the database design is well-structured and normalized up to BCNF.